



SPOTIFY

SQL DATA ANALYSIS

DATA ANALYSIS PROJECT

Reya Mondal

Data Analyst
SQL • Excel • Power BI



INTRODUCTION

Hello, I'm Reya Mondal

I am a Data Analyst with a strong interest in turning raw data into meaningful business insights. I work with SQL, Excel and Power BI to clean, analyze and present data in a clear and practical way.

PROFILE

Data Analyst Focused on analytical problem-solving

CORE SKILLS

SQL • Excel • Power BI
DAX • Power Query

PROJECT FOCUS

Spotify SQL Data Analysis
Data cleaning & business analysis

Goal: build reliable analysis and communicate insights clearly.

Project Overview

A SQL-driven exploration of Spotify track performance

BUSINESS GOAL

Understand streaming performance, artist activity, album insights, engagement and platform preference.

APPROACH

Clean the dataset → validate fields → aggregate metrics → compare performance → extract insights.

OUTPUT

10 business-oriented SQL analyses covering streams, artists, albums, audio features, views, likes and platform comparison.

Dataset & Table Structure

Main fields used in the analysis

Track & Artist

artist • track • album • album_type

Audio Features

danceability • energy • loudness •
speechiness • acousticness

Engagement

views • likes • comments

Streaming

stream • most_played_on

Video & License

licensed • official_video

Other Metrics

tempo • duration_min • valence •
liveness

Data Cleaning

Removing invalid duration records before analysis

1 CHECK ZERO-DURATION RECORDS

```
SELECT * FROM spotify  
WHERE duration_min = 0;
```

2 DELETE INVALID RECORDS

```
DELETE FROM spotify  
WHERE duration_min = 0;
```

QUALITY CHECK

Re-run the SELECT query after deletion to confirm that zero-duration records have been removed.

Analysis 1-3

Streaming, artist/album mapping and licensed engagement

01 • HIGH-STREAM TRACKS

```
SELECT *  
FROM spotify  
WHERE stream > 1000000000;
```

Identifies tracks crossing the 1-billion-stream threshold.

artist	track	album
Gorillaz	Feel Good Inc.	Demon Days
Red Hot Chili Peppers	Californication	Californication (Deluxe Edition)
Red Hot Chili Peppers	Under the Bridge	Blood Sugar Sex Magik (Deluxe Edition)
50 Cent	In Da Club	Get Rich Or Die Tryin
Coldplay	Yellow	Parachutes
Coldplay	Viva La Vida	Viva La Vida or Death and All His Friends
Coldplay	Something Just Like This	Memories...Do Not Open
Coldplay	The Scientist	A Rush of Blood to the Head
Coldplay	A Sky Full of Stars	Ghost Stories
Coldplay	Hymn for the Weekend	A Head Full of Dreams
Coldplay	Fix You	X&Y

Total rows: 385 Query complete 00:00:00.173 CRLF Ln 50, Col 26

02 • ALBUM + ARTIST

```
SELECT DISTINCT album, artist  
FROM spotify;
```

Creates a unique album–artist mapping.

album	artist
NEED A FAVOR	Jelly Roll
Agar Tum Na Hote (Original Motion Picture Soundtrack)	R. D. Burman
Algo De Mi	Conjunto Primavera
The Melodic Blue (Deluxe)	Baby Keem
A N N I V E R S A R Y	Bryson Tiller
Untouchables	Kom
Alright, Still (Bonus Track Version)	Lily Allen
Mozart: Requiem	Berliner Philharmoniker
#NAME?	Jesse & Joy
HUMBLE. (SKRILLEX REMIX)	Skrillex
Moondance (Deluxe Edition)	

Total rows: 14178 Query complete 00:00:00.126 CRLF Ln 54, Col 15

03 • LICENSED COMMENTS

```
SELECT SUM(comments)  
FROM spotify  
WHERE licensed = TRUE;
```

Measures total comments for licensed

total_comments
497015695

Showing rows: 1 to 1 Page 1

Result — Query 4

Count of total tracks by artist

04 • SINGLE TRACKS

```
SELECT track, album_type
FROM spotify
WHERE album_type = 'single';
```

Identifies tracks released as singles.

61	--4.Find all tracks that belong to the album type single.
62	select track
63	from spotify
64	where album_type='single';
65	
66	5.Count the total number of tracks by each artist

Data Output	Messages	Notifications
Showing rows: 1 to 1000		
	track character varying (255)	
1	New Gold (feat. Tame Impala and Bootie Brown)	
2	New Gold (feat. Tame Impala and Bootie Brown) - Dom Dolla Remix	
3	Cracker Island (feat. Thundercat)	
4	Best Friend - Remix	
5	Lux Æterna	
6	Screaming Suicide	
7	Get Lucky (feat. Pharrell Williams & Nile Rodgers) - Radio Edit	
8	Numb / Encore	
9	Deixa Ela Em Paz	
10	Pump It Louder	
Total rows: 4973		Query complete 00:00:00.137

Analysis 5

Track type, artist volume and audio characteristics

RESULT

The executed query returned 2,074 rows. The visible result shows artists such as Marisela, Ray Charles, Joey Bada\$\$, La Mosca Tse-Tse, TheFatRat, Leo Dan and Tyga with 10 tracks each.

05 • TRACKS BY ARTIST

```
SELECT artist, COUNT(track)
FROM spotify
GROUP BY artist;
```

Measures artist-level track volume.

	artist character varying (255)	total_track bigint
1	Marisela	10
2	Ray Charles	10
3	Joey Bada\$\$	10
4	La Mosca Tse-Tse	10
5	TheFatRat	10
6	Leo Dan	10
7	Tyga	10
8	Grupo Marea Registrada	10
Total rows: 2074		Query complete 00:00:00.134

Result — Query 6

Average danceability of tracks in each album

06 • AVG DANCEABILITY

```
72 --6.Calculate the average danceability of tracks in each album.
73     SELECT
74     album,
75     AVG(danceability) AS avg_danceability
76     FROM spotify
77     GROUP BY album;
```

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 1000

✎

🖨

Page No: 1

	album	avg_danceability
	character varying (255)	double precision
1	Energy (Expanded Edition)	0.64
2	Nursery Rhymes by CoComelon	0.78
3	Some	0.78
4	Acústico (Ao Vivo / Deluxe)	0.6057142857142858
5	The Bang Years 1966-1968 (The 23 Original Mono Recordings)	0.54
6	Aadukalam (Original Motion Picture Soundtrack)	0.595
7	Antidepresan	0.7
8	Love Dose: Benny Dayal	0.59
9	ANDRÓMEDA	0.67

Total rows: 11853

Query complete 00:00:00.193

RESULT

The executed query returned 11,853 album-level rows. The visible results include Energy (Expanded Edition) at 0.64, Nursery Rhymes by CoComelon at 0.78, Some at 0.78 and Acústico (Ao Vivo / Deluxe) at approximately 0.606. Evidence from executed SQL query

Analysis 7-9

Energy ranking, official video engagement and album views

07 • TOP 5 ENERGY

```
SELECT track, MAX(energy)
FROM spotify
GROUP BY track
ORDER BY max_energy DESC
LIMIT 5;
```

08 • OFFICIAL VIDEO

```
SELECT track, views, likes
FROM spotify
WHERE official_video = TRUE;
```

Reviews video engagement metrics.

09 • ALBUM VIEWS

```
SELECT album, SUM(views)
FROM spotify
GROUP BY album
ORDER BY total_views DESC;
```

Ranks albums by total views.

Analysis 10 — Spotify vs YouTube

Conditional aggregation + HAVING

BUSINESS QUESTION

Which tracks have more Spotify streams than YouTube streams?

SQL LOGIC

```
SELECT track,  
       COALESCE(SUM(CASE WHEN most_played_on='YouTube' THEN stream END),0) AS youtube_streams,  
       COALESCE(SUM(CASE WHEN most_played_on='Spotify' THEN stream END),0) AS spotify_streams  
FROM spotify  
GROUP BY track  
HAVING COALESCE(SUM(CASE WHEN most_played_on='Spotify' THEN stream END),0)  
       > COALESCE(SUM(CASE WHEN most_played_on='YouTube' THEN stream END),0);
```

SQL Concepts Demonstrated

Core techniques used throughout the project

01

SELECT / WHERE

02

DISTINCT

03

COUNT / SUM / AVG

04

GROUP BY

05

ORDER BY / LIMIT

06

CASE WHEN

07

COALESCE

08

HAVING

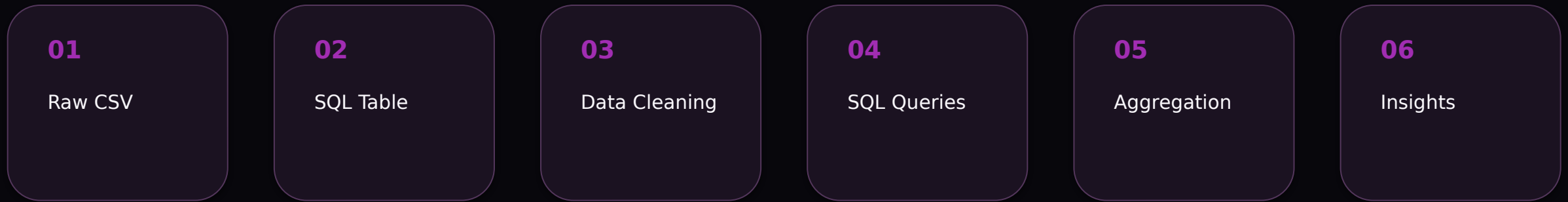
Key Analytical Takeaways

What this project demonstrates

- Streaming** Identify high-performing tracks using stream thresholds.
- Artists** Measure artist catalog volume with grouped counts.
- Albums** Compare album-level danceability and total views.
- Engagement** Use views, likes and comments as audience indicators.
- Platforms** Use conditional aggregation to compare platform-labelled streams.

Project Workflow

End-to-end SQL analytics process



Raw data → clean data → SQL analysis → grouped metrics → business-ready insights

SQL EXECUTION RESULTS

Actual PostgreSQL output screenshots added as project evidence

QUERY 5

2,074 result rows
Track count grouped
by artist

QUERY 6

11,853 result rows
Average danceability
grouped by album

PROJECT VALUE

Shows both the SQL
logic and the executed
database output

Use these result slides in your portfolio presentation to demonstrate that the SQL queries were actually executed.

THANK YOU

Spotify SQL Data Analysis

SQL • Data Cleaning • Data Analysis • Business Insights

Reya Mondal